

NABEEL REHEMTULLA

CURRICULUM VITAE

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EDUCATION

<i>PhD</i>	Northwestern University, Astronomy Presidential Fellow	<i>Expected 2027</i>
MS	Northwestern University, Astronomy Integrated Data Science Certificate	2023
BS	University of Michigan, Astronomy & Astrophysics with Highest Honors Second Major Interdisciplinary Physics; Minor Computer Science	2021

AWARDS AND FELLOWSHIPS

Outstanding Paper Award	2026
International Conference on Machine Learning – AI4Physics Workshop	
Presidential Fellowship	2025
Northwestern University – the <i>highest honor Northwestern bestows upon a graduate student</i>	
Outstanding Student Achievement Award	2024
Astroinformatics Conference, Puerto Natales, Chile	
2nd prize, Rapid Fire Research 2024	2024
Northwestern University, Department of Physics and Astronomy	
1st prize, Visualization Competition (animation)	2024
Northwestern University, Computation and Data Exchange (CoDEX)	
NASA Illinois Space Grant Consortium Graduate Fellowship (awarded twice)	2023, 2022
Highest Honors in Astronomy & Astrophysics	2021
University of Michigan, Department of Astronomy	
Chambliss Astronomy Achievement Award	2021
American Astronomical Society	
Outstanding Undergraduate Poster	2020
Michigan Institute for Data Science	

REFEREED PUBLICATIONS

Authored 30 papers (7 first-author, 3 major contribution) totaling >500 citations

FIRST-AUTHOR:

7. Li*, Chen*, **Rehemtulla***, et al. 2026, "*StarEmbed: Benchmarking Time Series Foundation Models on Astronomical Observations of Variable Stars*," Proceedings of the 43rd International Conference on Machine Learning, [arXiv:2510.06200](#)
*Equal contribution
6. **Rehemtulla**, Miller, Walmsley, et al., 2026, "*Pre-training vision models for the classification of alerts from wide-field time-domain surveys*," [PASP, 138, 034503](#)

5. **Rehemtulla**, Coughlin, Miller, et al. 2025b, "*The automation of optical transient discovery and classification in Rubin-era time-domain astronomy*," [NatAs, 9, 1764R](#)
4. **Rehemtulla**, Jacobson-Galán, Singh, et al. 2025a, "*The BTSbot-nearby Discovery of SN 2024jlf: rapid, autonomous follow-up probes interaction in an 18.5 Mpc Type IIP supernova*," [ApJ, 985, 241](#)
3. **Rehemtulla**, Miller, Jegou du Laz, et al., 2024, "*The Zwicky Transient Facility Bright Transient Survey. III. BTSbot: Automated Identification and Follow-up of Bright Transients with Deep Learning*," [ApJ, 972, 7](#)
2. **Rehemtulla**, Miller, Coughlin, et al., 2023, "*BTSbot: A Multi-input Convolutional Neural Network to Automate and Expedite Bright Transient Identification for the Zwicky Transient Facility*," Proceedings of the ML for Astrophysics Workshop at the 40th International Conference on Machine Learning, [arXiv:2307.07618](#)
1. **Rehemtulla**, Valluri & Vasiliev, 2022, "*Non-parametric spherical Jeans mass estimation with B-splines*," [MNRAS, 511, 5536](#)

MAJOR CONTRIBUTION:

3. Shah, **Rehemtulla**, Miller, et al., "*Leveraging Multimodality for Real-Time Classification of Transients and Variables found by the Zwicky Transient Facility*," 2026, arXiv:2607:00228
2. Gupta, Muthukrishna, **Rehemtulla**, et al., "*Transfer learning for transient classification: from simulations to real data and ZTF to LSST*," 2025, MNRAS, 542, L132
1. Medina, Li, Eadie, [incl. **Rehemtulla**] et al., "*The mass of the Milky Way from outer halo stars measured by DESI DR1*," 2025, arXiv:2508.19351

CO-AUTHOR:

20. Sevilla, Ho, Nayana, [incl. **Rehemtulla**] et al., "*Multiwavelength Analysis of Six Luminous, Fast Blue Optical Transients*," 2026, ApJ, 1007, 38S
19. Sasli, Xu, Junell, [incl. **Rehemtulla**] et al., "*Catching Disguised Transients with ASTRANet: Anomaly-Aware Spectroscopic Classification and Conformal Calibration*," 2026, arXiv:2607.08044
18. Simongini, Acero, Imbrogno, [incl. **Rehemtulla**] et al., "*Bridging the gap between SLSNe and SE-SNe. Multi-wavelength analysis of the SLSN-Ib SN 2024jlc*," 2026, arXiv:2607.01053
17. Kwok, Liu, Jha, [incl. **Rehemtulla**] et al., "*JWST Spectroscopy of SN Ia 2022aaq and 2024gy: Evidence for Enhanced Central Stable Ni Abundance and a Deflagration-to-Detonation Transition*," 2026, ApJ, 1005, 52K
16. Perley, Ho, McGrath, [incl. **Rehemtulla**] et al., "*AT2024wpp: An Extremely Luminous Fast Ultraviolet Transient Powered by Accretion onto a Black Hole*," 2026, MNRAS, 549, stag678
15. Junell, Sasli, Nunes, [incl. **Rehemtulla**] et al., "*Applying multimodal learning to Classify transient Detections Early (AppleCiDER) I: Data set, methods, and infrastructure*," 2026, PASP, 138, 054508

14. Singh, Kwok, Jha, [incl. **Rehemtulla**] et al., "*Photometry and Spectroscopy of SN 2024pxl: A Luminosity Link Among Type Iax Supernovae*," 2026, ApJ, 999, 227
13. Barjou-Delayre, Rosnet, Ravoux, [incl. **Rehemtulla**] et al., "*Study of the anisotropy of cosmic expansion on ZTF type Ia supernovae simulations*," 2026, arXiv:2601.11139
12. Burgaz, Maguire, Galbany, [incl. **Rehemtulla**] et al., "*ZTF SN Ia DR2 follow-up: Exploring the origin of the Type Ia supernova host galaxy step through Si II velocities*," 2026, A&A, 705, A76
11. Kasliwal, Ahumada, Stein, [incl. **Rehemtulla**] et al., "*ZTF25abjmnps (AT2025ulz) and S250818k: A Candidate Superkilonova from a Subthreshold Subsolar Gravitational-wave Trigger*," 2025, ApJ, 995, L59
10. Pearson, Subrayan, Sand, [incl. **Rehemtulla**] et al., "*Mid-infrared Dust Evolution and Late-time Circumstellar Medium Interaction in SN 2017eaw*," 2025, ApJ, 993, 213
9. Xu, Sasli, Junell, [incl. **Rehemtulla**] et al., "*AppleCiDEr II: SpectraNet -- A Deep Learning Network for Spectroscopic Data*," 2025, arXiv:2510.07215
8. Miller, Abrams, Aldering, [incl. **Rehemtulla**] et al., "*The La Silla Schmidt Southern Survey*," 2025, PASP, 137, 094204
7. Kwok, Singh, Jha, [incl. **Rehemtulla**] et al., "*JWST and Ground-based Observations of the Type Iax Supernovae SN 2024pxl and SN 2024vjm: Evidence for Weak Deflagration Explosions*," 2025, ApJ, 989, L33
6. Yao, Chornock, Ward, [incl. **Rehemtulla**] et al., "*A Massive Black Hole 0.8 kpc from the Host Nucleus Revealed by the Offset Tidal Disruption Event AT2024tvd*," 2025, ApJ, 985, L48
5. Das, Kasliwal, Fremling, [incl. **Rehemtulla**] et al., "*Low-luminosity Type IIP Supernovae from the Zwicky Transient Facility Census of the Local Universe. I. Luminosity Function, Volumetric Rate*," 2025, PASP, 137, 044203
4. Sharma, Mahabal, Sollerman, [incl. **Rehemtulla**] et al., "*CCSNscore: A Multi-input Deep Learning Tool for Classification of Core-collapse Supernovae Using SED-machine Spectra*," 2025, PASP, 137, 034507
3. Sharma, Sollerman, Kulkarni, [incl. **Rehemtulla**] et al., "*Dramatic Rebrightening of the Type-changing Stripped-envelope Supernova SN 2023aew*," 2024, ApJ, 966, 199
2. Strotjohann, Ofek, Gal-Yam, [incl. **Rehemtulla**] et al., "*Search for Supernova Progenitor Stars with ZTF and LSST*," 2024, ApJ, 960, 72
1. Cooper, Koposov, Allende Prieto, [incl. **Rehemtulla**] et al., "*Overview of the DESI Milky Way Survey*," 2023, ApJ, 947, 37

AWARDED TELESCOPE TIME AS PI

1.6 n; SOAR Telescope	NOIRLab 2026B
2.0 n; MMT Observatory	Northwestern 2026B
1.0 n; Gemini-N Observatory	NOIRLab 2026A
1.3 n; SOAR Telescope	NOIRLab 2026A
0.4 n; SALT Observatory	NOIRLab 2026A
2.5 n; MMT Observatory	Northwestern 2026A

1.0 n; Gemini-N Observatory	NOIRLab 2025B
3.5 n; MMT Observatory	Northwestern 2025B
1.0 n; SOAR Telescope	NOIRLab 2025A
3.0 n; MMT Observatory	Northwestern 2025A

PRESENTATIONS

INVITED TALKS AND SEMINARS:

Invited Lecture, Rise_Time 2026 – Precursors: Soiree for Rising Stars	Remote; Aug 2026
Seminar, Carnegie High Energy & Massive Stars Group Meeting	Pasadena, CA; May 2026
Invited Seminar, Caltech Special Seminar	Pasadena, CA; May 2026
Seminar, NOIRlab FLASH Seminar	Tucson, AZ; Apr 2026
Invited Seminar, Earth 2.0 Telescope Science Seminar	Remote; Jan 2026
Invited Talk, CCA Foundation Models for Astronomy	New York, NY; May 2025
Invited Talk, LS4 Team Meeting	Evanston, IL; Mar 2025
Invited Seminar, Rutgers University Astronomy Journal Club	Remote; Feb 2025
Invited Seminar, A3D3 Multi-messenger Astrophysics Monthly Telecon	Remote; Nov 2023

CONTRIBUTED TALKS AND POSTERS:

Talk, Empowering Science in the Data-rich Era of Astronomy	Tucson, AZ; Apr 2026
Talk, RAPID Response: Hotwiring Transients	Pasadena, CA; Mar 2026
Talk, TDABench 2026	Chicago, IL; Mar 2026
Talk, Astroinformatics 2026	Cologne, Germany; Feb 2026
Poster, Open SkAI 2025	Chicago, IL; Sep 2025
Talk & Panelist, Rutgers Summer Transient Soiree	Piscataway, NJ; Jul 2025
Talk, Celebrating 20 Years of Swift Discoveries	Florence, Italy; Mar 2025
Talk, 245 th AAS Meeting	National Harbor, MD; Jan 2025
Talk, Astroinformatics 2024	Puerto Natales, Chile; Dec 2024
Poster, Rise_Time 2024	West Lafayette, IN; Aug 2024
Poster, International Conference on Machine Learning	Honolulu, HI; Jul 2023
Talk, The Transient and Variable Universe	Urbana, IL; Jun 2023
Lecture, ASTRON 111 – Introduction to Astrobiology	Evanston, IL; Apr 2023
Poster, 241 st AAS Meeting	Seattle, WA; Jan 2023
Poster, 237 th AAS Meeting <i>DESI Special Session</i>	Remote; Jan 2021

PRESS

Total reach of >3.1M across >30 media mentions

Selected articles featuring "*SN 2023tyk: discovery to spectroscopic classification performed fully automatically*" (Rehemtulla et al. 2023)

- ◆ [Northwestern Now](#), “*First supernova detected, confirmed, classified and shared by AI*”
- ◆ [Space.com](#), “*AI just spotted its 1st supernova. Could it replace human explosion hunters?*”
- ◆ [Gizmodo](#), “*AI Detected a Supernova All on Its Own*”
- ◆ [Dropbox Blogs](#), “*An AI-powered telescope is helping astronomers sift through the stars*”

Selected articles featuring "*BTSbot: Automated Identification and Follow-up of Bright Transients with Deep Learning*" (Rehemtulla et al. 2024)

- ◆ [Northwestern Now](#), “*Sky survey classifies more than 10,000 supernovae*”
- ◆ [Caltech News](#), “*ZTF Leads to Classification of 10,000 Supernovae*”

SELECTED FIRST-AUTHOR ASTRONOTES

AstroNotes are circulars to the transient astronomy community

Rehemtulla, Jacobson-Galán, Kwok, et al., 2026, "Reclassification of SN 2026sqf, the nearest Type Ia-CSM to-date," TNSAN, 2026-240

Rehemtulla, Sollerman, Liu, et al., 2025b, "Discovery and classification of SN 2025ahqr, a highly reddened low luminosity Type Iax in M108 (9 Mpc)," TNSAN, 2025-345

Rehemtulla, Schulze, Jacobson-Galán, et al., 2025a, "Swift/UVOT observations of SN 2025qtt 6 minutes after its automatic discovery by BTSbot-nearby," TNSAN, 2025-209

Rehemtulla, Miller, Fremling, et al., 2023, "SN 2023tyk: discovery to spectroscopic classification performed fully automatically," TNSAN, 2023-265

- ◆ *5th most viewed AstroNote of all time*

MENTORING & LEADERSHIP

Northwestern University, CIERA, SkAI

Mentor, Satellite Early Education for Discovery (SEED) REU Program	Summer 2026
Committee Member, Graduate Student Council Mentorship Committee	2025 – 2026
Teaching Assistant, Code/Astro Workshop	Summer 2023
Teaching Assistant, ASTRON 111	Spring 2023
Teaching Assistant, ASTRON 120	Fall 2022
Mentor, Research Experiences at CIERA for High Schoolers	Summer 2022

Ismaili Center of Chicago (local mosque), Chicago, IL

Board Member, Midwest Professionals Network	Sep 2023 – Present
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